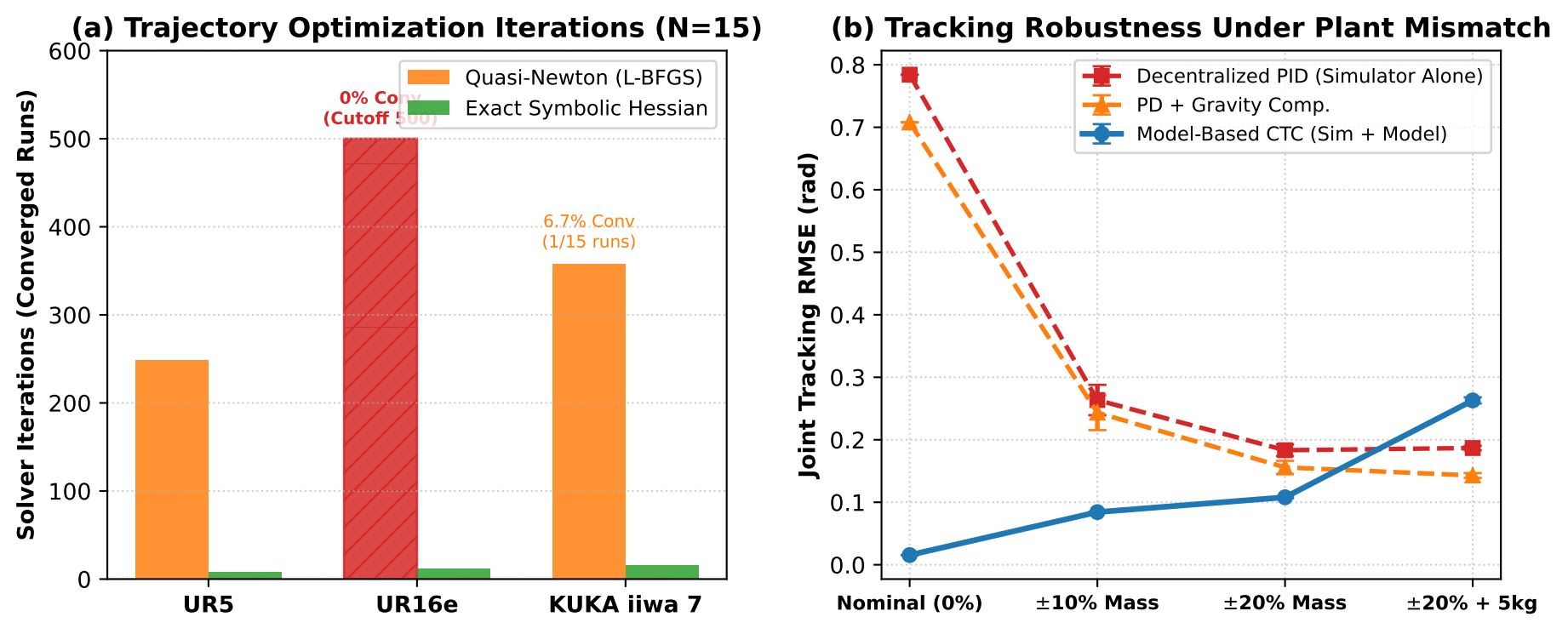
(a) Trajectory Optimization Iterations (N=15)
Solver Iterations (Converged Runs)
600
500
400
300
200
100
0
Quasi-Newton (L-BFGS)
Exact Symbolic Hessian
0% Conv
(Cutoff 500)
6.7% Conv
(1/15 runs)
UR5
UR16e
KUKA iiwa 7
(b) Tracking Robustness Under Plant Mismatch
Joint Tracking RMSE (rad)
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
Decentralized PID (Simulator Alone)
PD + Gravity Comp.
Model-Based CTC (Sim + Model)
Nominal (0%)
±10% Mass
±20% Mass
±20% + 5kg